

Screen Time, Two Deletions

AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7) · TEACHER KEY

CED TETHER

- VAR-1.A / UNC-1.A: identify the question to be answered and the individuals and variables in a context.
- VAR-1.B: classify variables as categorical or quantitative; a label or identifier is not a measurement.
- UNC-1.H / 1.I: compare groups of unequal size with relative frequencies, not raw counts.
- UNC-1.J: describe a distribution with shape, center, variability, and unusual features.
- UNC-1.K: the median is resistant to unusual values; the mean is not, so report the center that fits the data.

Essential question: What does a number mean, and what changes when you delete the values you do not like?

DOK-3 skill: 4.B · **FRQ pattern:** judge-a-trimmed-report-using-context-resistance-and-relative-frequency

DOK rationale: Part (d) requires a defended decision: whether a report that deleted two values is responsible, which statistic to report, and what the changed percent means for the principal’s actual question. It coordinates context, individuals-versus-variables, relative frequency, and resistance rather than recalling any one of them.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (d) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) and (c) are the ladder up; use them to see where a thin (d) came from.

Questions to ask

- Who exactly was measured, and what is one row of the data?
- Which of your variables could you add up, and which could you only count?
- If the two groups were the same size, would your comparison change?
- Which number moved a lot and which barely moved when the two values were deleted? Why?

[WARN] LOOK FOR

- Writing a number with no who/what/units (the 30% left floating).
- Calling the student ID or the device name a quantitative variable.
- Comparing 9 freshmen to 3 seniors by counts and declaring freshmen three times worse.
- Saying the mean is resistant, or that deleting the two values was fine because the median barely moved.
- Describing the distribution without mentioning the two unusual values at all.

SCORING PART (d) — E / P / I

Expected elements

- **e1** (required) Gives the 26% its context: who (students in the sample), what (daily screen time over 240 minutes), and the question it answers.
- **e2** (required) — Chooses a center and justifies it with resistance, consistent with part (c).
- **e3** (required) — Interprets the big mean change versus the small median change as the effect of the two unusual values.
- **e4** (required) — States that the deletion must be disclosed and justified (logging errors) and that unusual values still belong in the description.

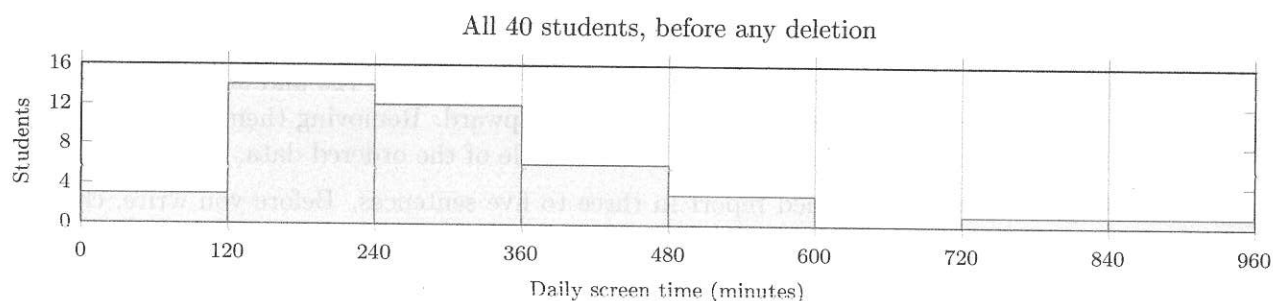
E	All four elements, in the context's words, with a clear decision
P	A decision with two or three elements, or all four with the context or the disclosure vague.
I	A bare number or a decision without resistance, context, or the disclosure.

[STAR] THE PROBLEM — annotated

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*



Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

Key: Ungraded commitment; accept any position that names something the first sentence leaves out.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level		
Device used most		
Daily screen time		

Key: Individuals: the 40 students in the sample. Grade level: one value is Freshman or Senior, categorical. Device used most: one value is Phone, Laptop, or Console, categorical. Daily screen time: one value is a number of minutes such as 176, quantitative (it can be averaged).

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the _____ freshmen, _____ are over 240, which is _____%. Of the _____ seniors, _____ are over 240, which is _____%. So the fair comparison is _____ because the groups _____.* Then explain in one sentence why the count 9 versus 3 can mislead here.

Key: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. The fair comparison is 36% versus 20% because the groups have different sizes; 9 versus 3 exaggerates the gap (three times as many students, not three times as bad), and freshmen are still higher but by less than the counts suggest.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

Key: The median is resistant; the mean is not. The two unusual values at 720 and 900 minutes sit far to the right of the rest of the distribution, so they pull the mean upward. Removing them drops the mean by 27 minutes, while the median, which depends only on the middle of the ordered data, moves by 2.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) **WHIC**, **WHAT**, and the **UNITS** behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must **SAY** about them. Use the frame below if it helps.

Key: Report the median (about 174–176 minutes) because it is resistant and describes the typical student either way. Removing the two values changed the mean a lot but barely changed the median, which shows the mean was being pulled by two unusual values, not by typical students. The 26% means that 10 of the 38 remaining students in the sample reported more than 240 minutes per day. Deleting the two entries is defensible because they were logging errors, not real days, but the report must say that two values were removed and why, and it should still note that unusual values existed; only then does the number connect to the principal's question about whether screen time is a concern.

NAME: Tom NguyenPERIOD: F**Screen Time, Two Deletions**

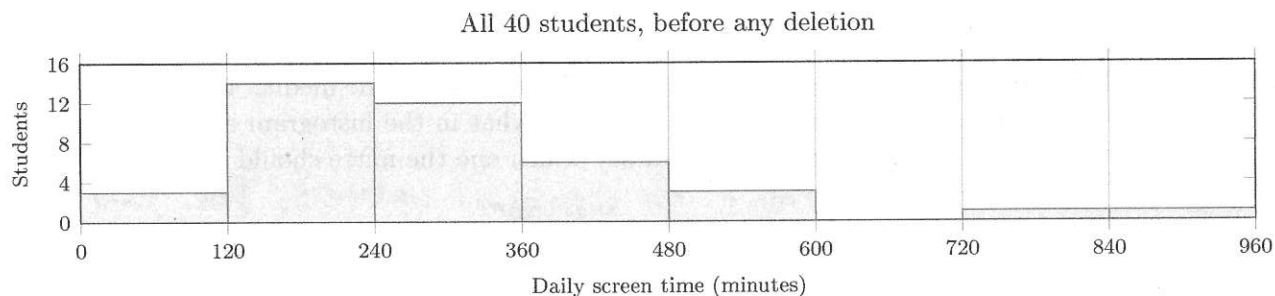
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[STAR] THE PROBLEM

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Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

Yes, however the nurse leaves out the screen time and the student labels.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level	12th Senior	Categorical
Device used most	Phone	Categorical
Daily screen time	240 minutes	Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is freshmen because the groups bigger.* Then explain in one sentence why the count 9 versus 3 can mislead here. *Since freshmen have a larger sample size we need to compare the percentages proportional to the amount of people not the count.*

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet. *The median is more resistant to unusual values. The two large outliers caused the histogram's mean to rise.*

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I would report that 26% of students have a daily screen time above 240 minutes. 26% does mean yes, screen time does affect students. Also deleting the two outliers does make sense, since they mistakenly reported a week's worth of screen time.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Darla GonzalezPERIOD: E**Screen Time, Two Deletions**

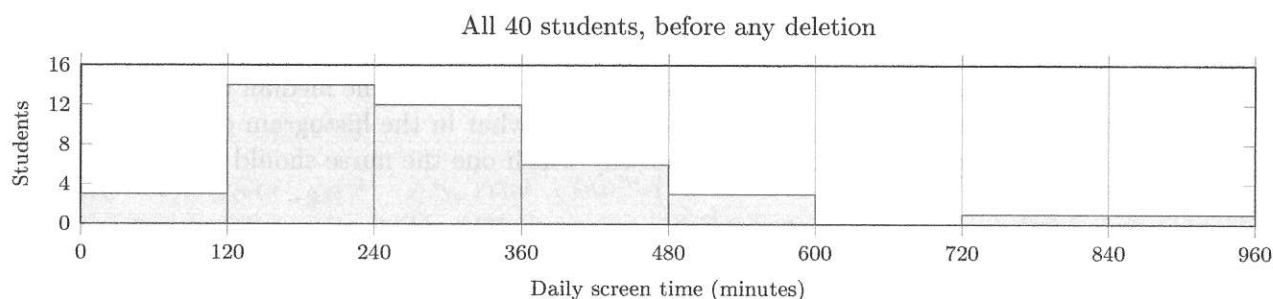
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[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	110 students	
Variable	One value looks like	Categorical / Quantitative
Grade level	freshman	Categorical
Device used most	Phone	Categorical
Daily screen time	240 min	Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36 because the groups are different sizes. Then explain in one sentence why the count 9 versus 3 can mislead here.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.
The median is resistant to unusual values. The unusual values were 120 min and 900 min. When those unusual values were removed, the mean changed by 27 min.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

Due Monday

NAME: Jessely Ampudia

PERIOD: E

Screen Time, Two Deletions

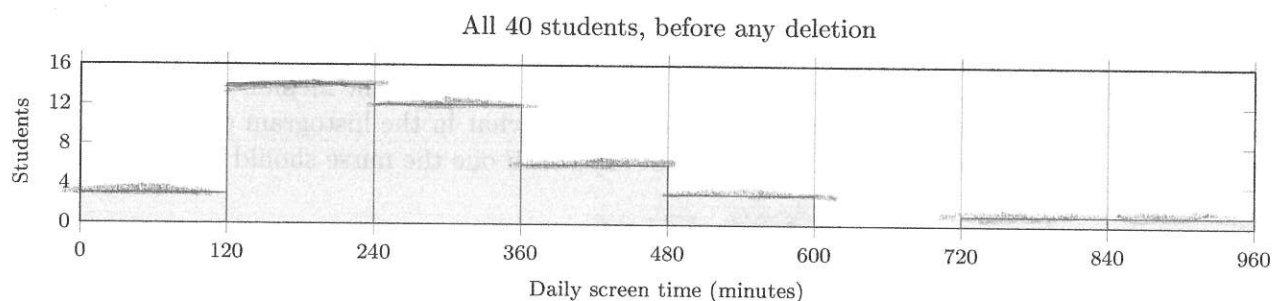
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[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*



FIRST TAKE — one sentence before you work the parts

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

The nurse left out the "too much" screen time.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	40 students	
Variable	One value looks like	Categorical / Quantitative
Grade level	Freshman	Categorical
Device used most	Phone	categorical
Daily screen time	120 min	Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36% vs 20% because the groups ^{sizes} have different. Then explain in one sentence why the count 9 versus 3 can mislead here.

It doesn't account for the different group sizes.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

I don't get it

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I don't get it

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Damian CepedaPERIOD: E**Screen Time, Two Deletions**

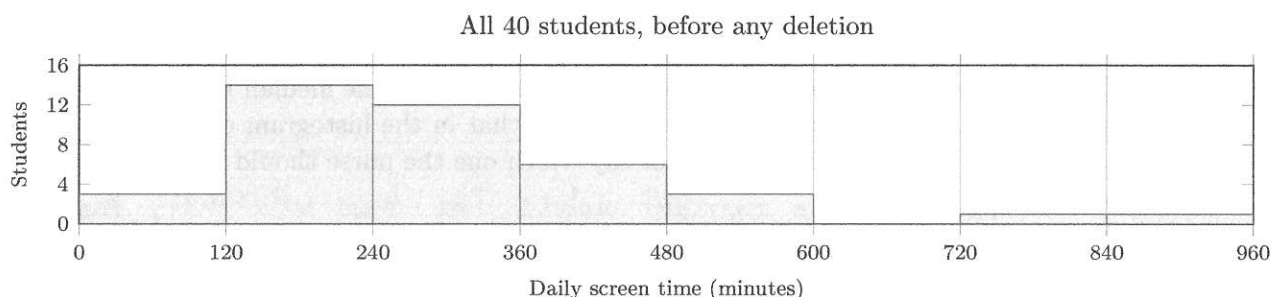
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Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

No. It does not explain why being over 240 minutes means screen time is a concern.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	The 40 students	
Variable	One value looks like	Categorical / Quantitative
Grade level	Freshman	Categorical
Device used most	Phone	Categorical
Daily screen time	180 minutes	Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36% vs 20% because the groups differ.* Then explain in one sentence why the count 9 versus 3 can mislead here.

The count 9 versus 3 can be misleading because there are more freshmen than seniors.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is resistant to unusual values. The two unusually high values pulled the mean up a lot while the median barely changed.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I would report the median of 174 minutes because it is resistant to unusual values. The 26% means 10 of the 38 students had over 240 minutes of daily screen time but that does not fully answer whether screen time is a concern. Removing the two values was reasonable because they were mistakenly recorded for a whole weekend but the report should say they were removed.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Mr. CokerPERIOD: B**Screen Time, Two Deletions**

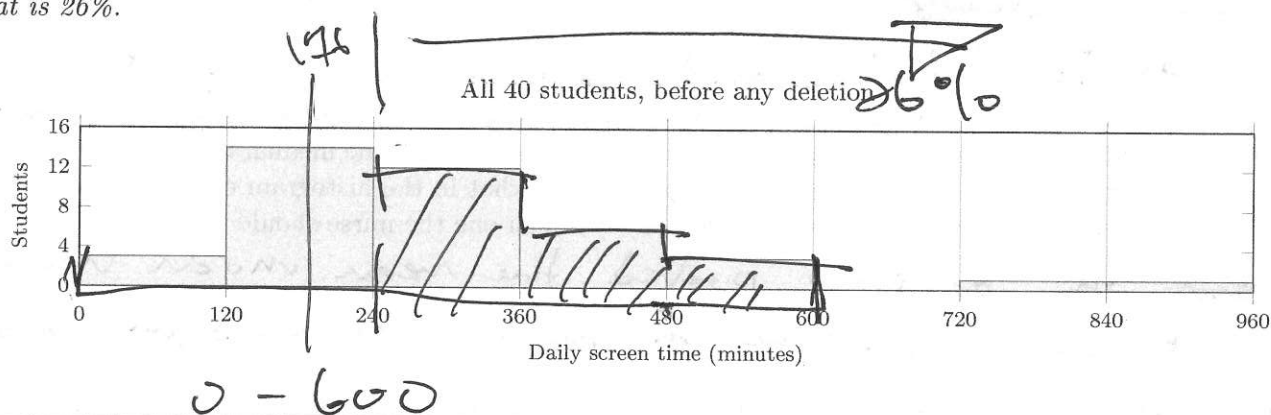
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[STAR] THE PROBLEM

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Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

30% what? 240 min / day no units

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)-(d).

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Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level	freshman	Cat.
Device used most	phone	Cat
Daily screen time	60	Quant

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the ~~34~~ freshmen, 9 are over 240, which is 26%. Of the ~~15~~ seniors, 3 are over 240, which is 20%. So the fair comparison is missing because the groups change. Then explain in one sentence why the count 9 versus 3 can mislead here.

20% < 26%, sounds as bad but nowhere near as hyperbolic as ques 3 (frms group totals)

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

the outliers pushed the mean much higher than the median, so we say median is resistant to outliers.

DOK 3 (d) * Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

Freshman are represented more than seniors. 40 students, median was 174 minutes a day. The median of a sample of 40 high school students' screen time usage yielded a median of

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

174 minutes a day with no outliers.

26% of our sample (most majority freshmen 5:3) was above the 240 minute / day cutoff.

NAME: Colleen FretiPERIOD: B**Screen Time, Two Deletions**

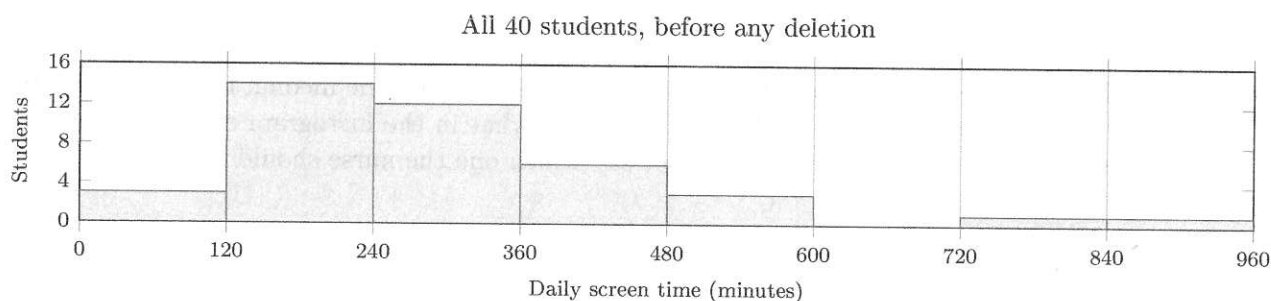
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**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

Nope the nurse's statement doesn't mean much since we don't know what that screen time does to students not only that the statistic is a stark minority

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level	9th grade	categorical
Device used most	Samsung	categorical
Daily screen time	80000 minutes	quantitative

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is misleading because the groups differ. Then explain in one sentence why the count 9 versus 3 can mislead here.

The sentence misleads because the total percent differs the senior judged only the numbers and not total

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is resistant to unusual values that range from 20 to 960 this shifted the mean toward

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

The 26% are the students who's screen time range from 0 - 600 minutes the percentage acknowledges the all students not including the 2 over 600 this doesn't answer the

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Elmer Lucero y Lucero

PERIOD: _____

Screen Time, Two Deletions

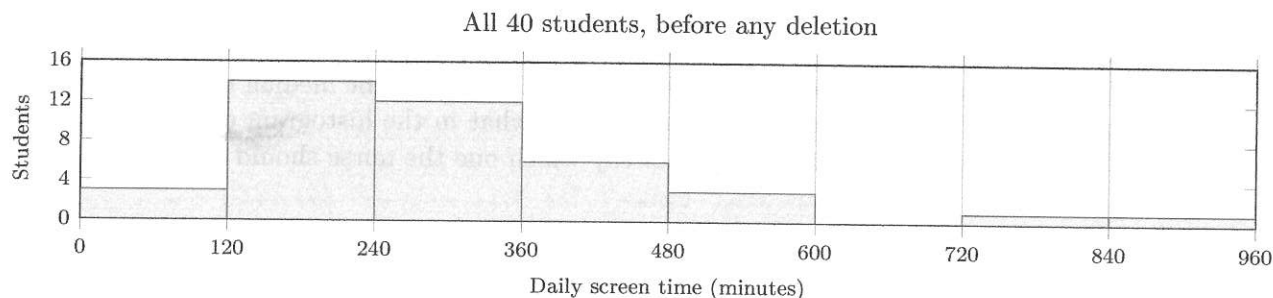
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

The nurse answers the question somewhat, as the nurse gives data first than giving the answer, then the explanation.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	Student screen time	
Variable	One value looks like	Categorical / Quantitative
Grade level	Freshman, Senior	Categorical
Device used most	Phone, laptop, console	Categorical
Daily screen time	over 240 min	Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 18 seniors, 3 are over 240, which is 16.7%. So the fair comparison is to use 9 and 3 because the groups have different sizes. Then explain in one sentence why the count 9 versus 3 can mislead here.

Comparing 9 versus 3 is misleading because there's more freshmen than seniors, making it unequal.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is the resistant to unusual values. The presence of extreme outliers is what caused the mean to move, while the median was stable.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps. I would report the median because it resists the influence of the two extreme outlier values, but not the mean, so the principal should keep the data; removing the two values, proving those values were the reason for the distortion in the dataset while the median remained the same.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Sally UneyPERIOD: B**Screen Time, Two Deletions**

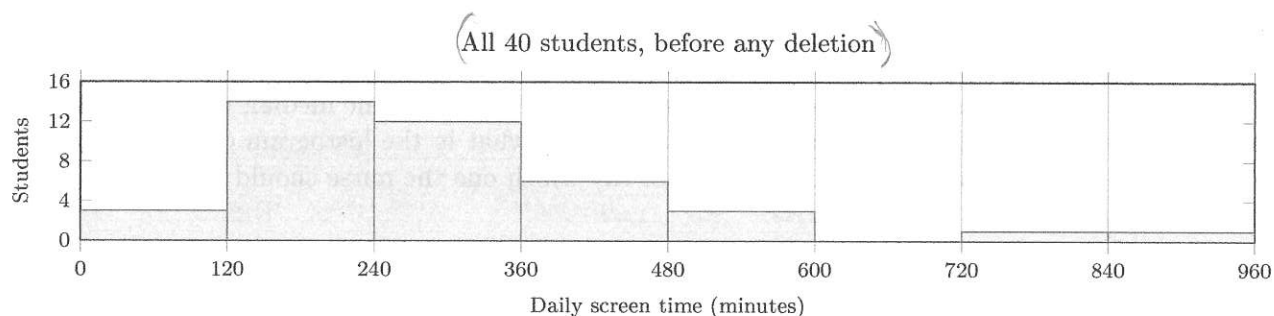
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: Is too much screen time a concern for our students? The nurse's first report said only: 12 of the 40 are over 240, that is 30%, so yes

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became 10 of 38 are over 240, that is 26%.

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

No. It leaves out what screen time does making the data meaningless.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	<u>student</u>	
Variable	One value looks like	Categorical / Quantitative
Grade level	<u>1st Grade</u>	<u>Categorical</u>
Device used most	<u>Nokia</u>	<u>Categorical</u>
Daily screen time	<u>67 minutes</u>	<u>Quantitative</u>

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is misleading because the groups differ. Then explain in one sentence why the count 9 versus 3 can mislead here.

The totals differ so you cannot judge by numbers. You can only judge by proportions or parts of a whole.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is resistant to unusual values. There is a interval ranging from 720-960 which shifted the mean significantly from 205-178.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

Behind the 26% are all the students with screen times ranging from 0 to 600 but not exceeding. The 26% answers for the second half minus outliers above 240, which does not answer as

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

B Due 9/21

NAME: Yolondys Garcia Reyes

PERIOD: B

Screen Time, Two Deletions

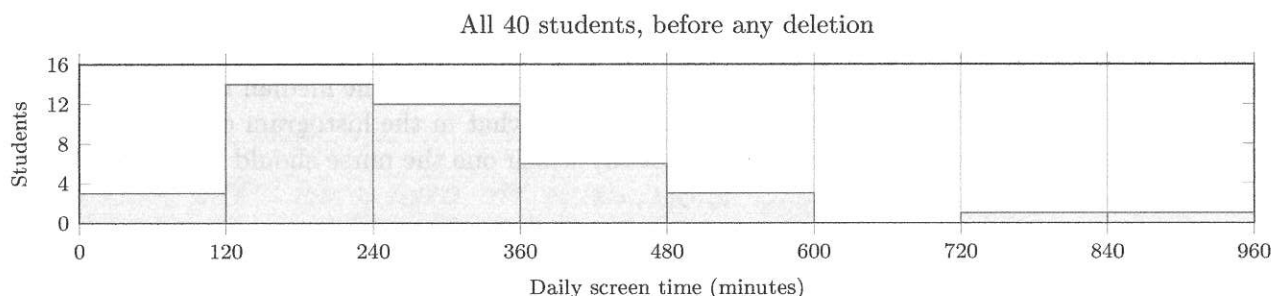
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*



FIRST TAKE — one sentence before you work the parts

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level	Freshman or senior	Categorical
Device used most	Phone, Laptop, or Console	Categorical
Daily screen time		Quantitative

DOK 2 (b) A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36% vs 20% because the groups are different size.* Then explain in one sentence why the count 9 versus 3 can mislead here. *The count 9 versus 3 can mislead because the groups have different sizes.*

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is resistant to unusual values, while the mean is not. The two unusually large values of 780 and 900 minutes pulled the mean upward, but the mean changed by only 2 minutes.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I would report that 10 of the 38 students (26%) reported over 240 minutes of daily screen time because it describes the students after the two mistaken weekend entries were removed. After the two values were removed the mean changed by 27 minutes and the median only by 2 minutes, showing that the median is resistant to unusual values. The 26% answers how many students reported over 240 minutes, not whether screen time is usually a concern. The principal should understand that the results are based on student's self-reporting and that the two entries were removed because they were added as a mistake.

Frame: I would report 26% because 240 minutes; removing the two values changed mean by 27 minutes but not the median, so the principal should consider the results carefully because they are self-reported and based on a small sample. *changed by only 2 minutes.*

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Zabby cleverPERIOD: B**Screen Time, Two Deletions**

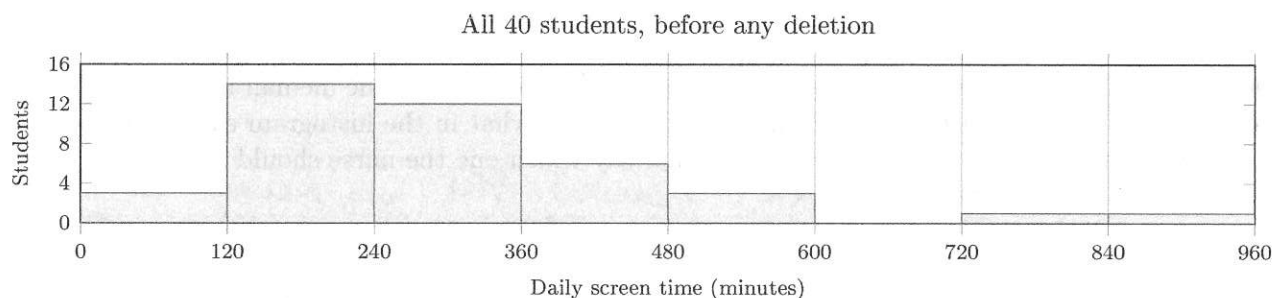
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

It leaves out what the data means.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

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Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	A randomly selected student	
Variable	One value looks like	Categorical / Quantitative
Grade level	senior	categorical
Device used most	phone	categorical
Daily screen time	240 minutes	quantitative

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the 24 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36 because the groups freshmen. Then explain in one sentence why the count 9 versus 3 can mislead here.

It can mislead because the two groups have unequal sample sizes.

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The unusual values caused the big move in the histogram. The median is resistant because it only depends on the relative ordering of the data points.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I would report the median because it's resistant to unusual values; removing the two values changed the mean but not the median, so the principal should use the median of 174 minutes.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.

NAME: Rubiat RahmanPERIOD: B**Screen Time, Two Deletions**

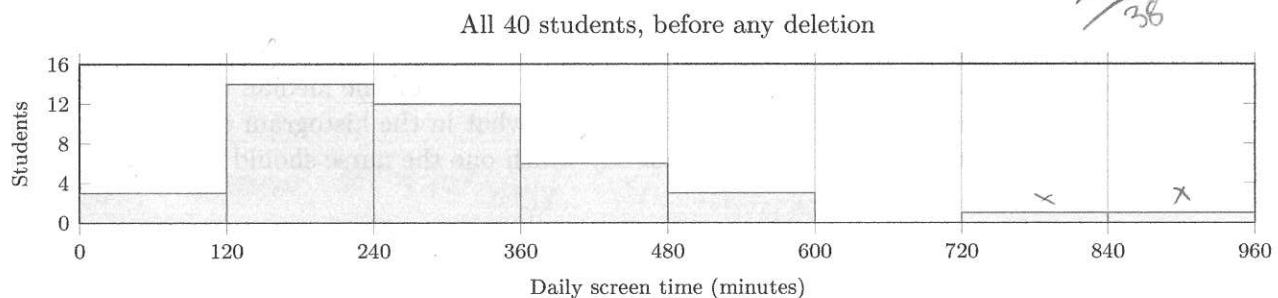
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out. *NO, because we don't know what 'too much' mean. The sentence leave out labeling 240 minutes*

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	40 student	
Variable	One value looks like	Categorical / Quantitative
Grade level	freshmen	categorical
Device used most	Phone	categorical
Daily screen time	120 minutes	Quantitative

$$\frac{25}{9} =$$

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is 36% compared to 20% because the groups sample size. Then explain in one sentence why the count 9 versus 3 can mislead here.
Comparing row counts is misleading here because the n number of freshmen and seniors is different. total

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is statistic is resistant to unusual values because the mean factors every value while the median only tracks the middle data position.

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

I would report the median because median is resistant to unusual values. Removing the two values changed mean by alot but not the median, so the principal should given the median report.

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.